

8	(a) (i)	In photoelectric effect, <u>EM radiation releases electrons</u> . In X-ray production, the reverse happens whereby <u>electrons' acceleration or deceleration releases EM radiation</u> .	B1
	(ii)	Maximum $I = P/V = 4 \times 10^3 / 40 \times 10^3 = 0.10 \text{ A} = 100 \text{ mA}$ Minimum $I = P/V = 4 \times 10^3 / 115 \times 10^3 = 0.0348 \text{ A} = 35 \text{ mA}$ Range of currents = 35 – 100 mA	M1 A1
	(iii)	The rest of the <u>kinetic energy of the electrons is transferred to the atoms</u> (which causes the atoms to vibrate more vigorously). Or The rest of the kinetic energy is <u>converted to heat</u> .	B1 B1
	(b)(i)	310	B1
	(ii)	Accurate plot (to half a small square) Best fit line	B1 B1

	(iii)	Gradient = $(3.12 - 0.12) / (1640 - 40) = 1.875 \times 10^{-3} \text{ keV} \approx 1.9 \times 10^{-3} \text{ keV}$ Thus, $aE_1 = 1.9 \times 10^{-3} \text{ keV}$ $\Rightarrow E_1 = 1.9 \times 10^{-3} \text{ keV} / 0.139 = 13.7 \text{ eV}$ Or $E_1 = 1.875 \times 10^{-3} \text{ keV} / 0.139 = 13.5 \text{ eV}$	M1 A1
	(iv)	Finding y-intercept: $3.12 = (1.9 \times 10^{-3})(1640) + c \Rightarrow c = 0.0040 \text{ keV}$ Or $3.12 = (1.875 \times 10^{-3})(1640) + c \Rightarrow c = 0.045 \text{ keV}$ Since $c \approx 0 \text{ keV}$, this is consistent with the mathematical expression of Moseley's law which does not have a y-intercept. (Alternatively, students can read off the y-intercept and comment on the consistency.)	M1 A1
	(v)	For the K_α line, $E = 0.75E_1(Z - 1)^2 = 0.75(0.0137)(Z - 1)^2$ Thus, $E = 0.75(0.0137)(74 - 1)^2 = 54.8 \text{ keV}$	M1 A1
	(vi)	To show up on the X-ray spectrum of rhodium, the minimum energy of the incident electrons must be 2.70 keV. Thus, the minimum accelerating voltage must be 2.70 kV.	M1 A1
	(c)(i)1	$\frac{\text{Intensity of X-ray beam emerging from model}}{\text{Intensity of X-ray beam incident on model}} = \frac{I_0 e^{-(0.3)(10)}}{I_0}$ $= 0.0498$	A1
	(i)2	$\frac{\text{Intensity of X-ray beam emerging from model}}{\text{Intensity of X-ray beam incident on model}} = \frac{I_0 e^{-(0.3)(4)} e^{-(10)(6)}}{I_0}$ $= 2.64 \times 10^{-27}$	M1 A1
	(ii)	The ratio of the ratios in (i) and (ii) is very large. Thus, the images will have good contrast	M1 A1